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Optimal Feedback Controls in Dynamic Stochastic Jobshops

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ABSTRACT. We consider a production planning problem for a general jobshop subject to breakdown and repair of machines and subject to lower and upper bound constraints on work-in-process. The machine capacities and demand processes are assumed to be finite state Markov chains. The problem is to choose the rates of production on the various machines over time to minimize the expected discounted costs of production and inventory/backlog over an infinite horizon. It is formulated as a stochastic dynamic programming problem. We show that the value function of the problem is locally Lipschitz and is a solution to a dynamic programming equation together with a certain boundary condition. We provide an interpretation of the boundary condition, provide a verification theorem, and derive the optimal feedback control policy in terms of the directional derivatives of the value function. The results are proved via reduction to a deterministic optimal control problem that is equivalent to the stochastic production planning problem under consideration.

1. Introduction

We consider a manufacturing system producing a variety of products in demand using machines in a general network configuration, which generalizes both the parallel and the tandem machine models. Each product follows a process plan—possibly from a number of alternative process plans—that specifies the sequence of machines it must visit and the operations performed by them. A process plan may call for multiple visits to a given machine, as is the case in semiconductor manufacturing (Lou and Kager [**LK**], Srivatsan, Bai, and Gershwin [**SBG**], Uzsoy, Lee, and Martin-Vega [**ULM1, ULM2**], and Yan et al. [**YLSG**]). Often the machines are unreliable. Over time they break down and must be repaired. A manufacturing system so described will be termed a *dynamic jobshop*. The term will be made mathematically precise in the next section.

The problem we are interested in is that of obtaining the optimal feedback rates of production of intermediate parts and of finished products in a manufacturing system consisting of a network of failure-prone machines and limited buffers between the machines. The objective is to meet demand for finished products at the minimum possible total discounted cost of production, inventories, and backlogs. It is essential for our analysis to have a precise specification of the general dynamics representing a jobshop. Sethi and Zhou [**SZh**] have established a graph-theoretic

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final products, $u_{ij}(t)$ is the rate of conversion from the material in buffer i to the material in buffer j by using the machine on the arc (i,j), $i = 0,1,2,3$ and $j = 1,2,3,4,5$. Let $x_i(t)$, $i = 1,...,5$, be the contents of buffer b_i at time t. Clearly $x_i(t)$ for $i = 1,2,3$, cannot become negative, and for $i = 4,5$ it defines the inventory/shortage of final product i and can take negative and positive values.

The system dynamics and the constraints are stated below.

System dynamics:

$$\begin{aligned} &\dot{x}_1(t) = u_{01}(t) - u_{12}(t) - u_{14}(t), && \dot{x}_4(t) = u_{14}(t) + u_{34}(t) - z_4(t),\\ &\dot{x}_2(t) = u_{12}(t) - u_{23}(t), && \dot{x}_5(t) = u_{05}(t) - z_5(t);\\ &\dot{x}_3(t) = u_{23}(t) - u_{34}(t), && \end{aligned} \tag{2.1}$$

State constraints:

$$0 \le L_i \le x_i(t) \le H_i,\ i = 1,2,3 \quad \text{and} \quad -\infty < x_i(t) \le H_i,\ i = 4,5; \tag{2.2}$$

Control constraints:

$$\begin{aligned} &p_{01}u_{01}(t) \le k_1(t), && p_{05}u_{05}(t) + p_{14}u_{14}(t) \le k_4(t),\\ &p_{12}u_{12}(t) + p_{34}u_{34}(t) \le k_3(t), && p_{23}u_{23}(t) \le k_2(t), \end{aligned} \tag{2.3}$$

where H_i is the size of the buffer $i, i = 1,2,3,4,5$, L_i is the lower bound on the surplus in internal buffer $i, i = 1,2,3$, and p_{ij} represents required machine capacity for unit production rate of part j from part i.

A general jobshops of the type shown in Fig. 1 and described by (2.1)-(2.3) can be defined as follows. Note immediately that the placement of machines in Fig. 1 plays no role in describing the dynamics (2.1) and constraints (2.2). Therefore, to describe the jobshop dynamics, we can remove machines from Fig. 1 and consider only the digraph whose vertices correspond to the buffers and whose arcs correspond to flows between the connected buffers as shown in Fig. 2.

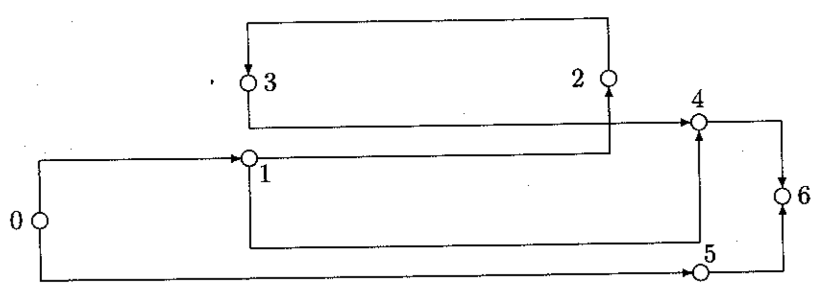

Fig. 2. The Digraph Corresponding to Fig. 1

The placement of machines, which can be expressed by a *partition* of the set of the arcs of the digraph, supplies us with the control constraints (2.3). Note that a different placement of a possibly different number of machines (i.e., a different partition) would result in a different set of control constraints, and thus a different jobshop; see Fig. 3 for an example.

This observation enabled Sethi and Zhou to describe a general dynamic jobshop of interest by a digraph with partition.

THEOREM 2.4. *We can label all the vertices from* 0 *to* $N+1$ *in a way so that the label numbers of the vertices along every path are in a strictly increasing order, the source is labeled* 0, *the sink is labeled* $N+1$, *and the external buffers are labeled* $m+1, m+2, \dots, N$.

PROOF. The proof is similar to Theorem 2.2 of Sethi and Zhou **[SZh]** and is omitted. $\square$

With the help of Theorem 2.4, one is able to formally write the dynamics and the state constraints associated with a given manufacturing digraph. But first let us give a few definitions.

DEFINITION 2.5. For each arc (i,j), $j \neq N+1$, in a manufacturing digraph, the rate at which parts in buffer i are converted to parts in buffer j is labeled as *control* u_{ij}. Moreover, the control u_{ij} associated with the arc (i,j) is called an *output* of i and an *input* to j. In particular, outputs of the source are called *primary controls* of the digraph. For each arc $(i, N+1)$, $i = m+1, \dots, N$, the demand for products in buffer i is denoted by z_i.

In what follows, we shall also set

$$\begin{aligned} u_{i,N+1} &= z_i, \ i = m+1, \dots, N, \\ u_{ij} &= 0 \ \text{ for } (i,j) \notin A, \ 0 \le i \le N, \ 1 \le j \le N+1 \end{aligned}$$

in the interest of a unified notation. While z_i and u_{ij}, for $(i,j) \notin A$, are not controls, we shall for convenience refer to u_{ij}, $0 \le i \le N, 1 \le j \le N+1$, as controls. In this way, we can consider the controls as an $(N+1) \times (N+1)$ matrix $\mathbf{u} = (u_{ij})$ of the form shown in Fig. 4. The set of all such controls is written as $\mathcal{U}$, i.e.,

$$\mathcal{U} = \{\mathbf{u} = (u_{ij}) : 0 \le i \le N, 1 \le j \le N+1, u_{ij} = 0 \text{ for } (i,j) \notin A\}.$$

$$\begin{array}{ccccccccccc}
u_{0,1} & u_{0,2} & \cdots & u_{0,i} & u_{0,i+1} & \cdots & u_{0,m+1} & u_{0,m+2} & \cdots & u_{0,N} & 0 \\
0 & u_{1,2} & \cdots & u_{1,i} & u_{1,i+1} & \cdots & u_{1,m+1} & u_{1,m+2} & \cdots & u_{1,N} & 0 \\
\cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\
0 & 0 & \cdots & u_{i-1,i} & u_{i-1,i+1} & \cdots & u_{i-1,m+1} & u_{i-1,m+2} & \cdots & u_{i-1,N} & 0 \\
0 & 0 & \cdots & 0 & u_{i,i+1} & \cdots & u_{i,m+1} & u_{i,m+2} & \cdots & u_{i,N} & 0 \\
\cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\
0 & 0 & \cdots & 0 & 0 & \cdots & u_{m,m+1} & u_{m,m+2} & \cdots & u_{m,N} & 0 \\
0 & 0 & \cdots & 0 & 0 & \cdots & 0 & 0 & \cdots & 0 & z_{m+1} \\
\cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\
0 & 0 & \cdots & 0 & 0 & \cdots & 0 & 0 & \cdots & 0 & z_{N+1}
\end{array}$$

Fig. 4. The structure of the control $\mathbf{u}$

Now we shall write the dynamics and the state constraints corresponding to a manufacturing digraph (V, A) containing $N+2$ vertices consisting of a source, a sink, m internal buffers, and $N-m$ external buffers associated with $N-m$ distinct final products to be manufactured. We label all the vertices according to Theorem 2.2. For simplicity in the sequel, we shall call the buffer whose label is i as *buffer* i, $i = 1, 2, \dots, N$. We denote the surplus at time t in buffer i by $x_i(t)$, $i \in V \setminus \{0, N+1\}$, and the control at time t associated with arc (i,j) by $u_{ij}(t)$, $(i,j) \in A$.

where we have assumed that the required machine capacity p_{ij} (for unit production rate of part type j from part type i) equals 1, for convenience in exposition; see (2.3). Moreover, the corresponding surplus process satisfies constraints (2.5).

The analysis in this paper can be readily extended to the case when p_{ij} is any given positive constant.

Before proceeding to the next section, it may be convenient to recapitulate the following system parameters that we have used:

m_c : number of machines;
m : number of internal buffers;
r : number of arcs in the manufacturing digraph; and
$N-m$: number of external buffers.

For example, in Fig. 1, $m_c = 4$, $m = 3$, $r = 8$, and $N = 5$. In the particular case of flowshop, $m_c = r = N+1$ and $m = N$.

It is convenient also to introduce some notations. For $\mathbf{u} = (u_{ij}, 0 \leq i \leq N, 1 \leq j \leq N+1)$, we denote by $U_i^1(\mathbf{u})$ the total input and by $U_i^2(\mathbf{u})$ the total output of buffer i, $1 \leq i \leq N$, i.e.,

$$U_i^1(\mathbf{u}) = \sum_{l=0}^{N} u_{li} = \sum_{l=0}^{(i-1)\wedge m} u_{li}, \quad 1 \leq i \leq N,$$

$$U_i^2(\mathbf{u}) = \sum_{l=1}^{N+1} u_{il} = \begin{cases} \sum_{l=i+1}^{N} u_{il}, & 1 \leq i \leq m, \\ u_{i,N+1} = z_i, & m+1 \leq i \leq N. \end{cases}$$

With these notations the state dynamics (2.4) can be rewritten as

$$\dot{x}_i(t) = U_i^1(\mathbf{u}(t)) - U_i^2(\mathbf{u}(t)), \quad i = 1, \ldots, N. \tag{2.7}$$

For later use we also write equation (2.7) as

$$\dot{\mathbf{x}}(t) = \mathcal{A}\mathbf{u}(t),$$

and it is obvious that the operator $\mathcal{A}$ is linear.

3. The optimization problem

We are now in the position to formulate our stochastic optimal control problem for the jobshop defined by (2.4), (2.5), and (2.6). To formulate the constraints precisely, let

$$S = \prod_{i=1}^{m} [L_i, H_i] \times \prod_{i=m+1}^{N} (-\infty, H_i],$$

and for $\mathbf{k} = (k_1, \ldots, k_{m_c}, z_{m+1}, \ldots, z_N)$,

$$U(\mathbf{k}) = \Big\{\mathbf{u} = (u_{ij}), \mathbf{u} \in \mathcal{U},\ 0 \leq \sum_{(i,j)\in K_n} u_{ij} \leq k_n, \quad 1 \leq n \leq m_c, u_{i,N+1} = z_i,\ m+1 \leq i \leq N\Big\}, \tag{3.1}$$

and for $\mathbf{x} \in \mathbb{R}^N$ and $\mathbf{k}$,

$$U(\mathbf{x},\mathbf{k}) = \{\mathbf{u} : \mathbf{u} \in U(\mathbf{k}); x_i = L_i \Rightarrow U_i^1(\mathbf{u}) - U_i^2(\mathbf{u}) \geq 0,\ i = 1, \ldots, m,$$
$$x_i = H_i \Rightarrow U_i^1(\mathbf{u}) - U_i^2(\mathbf{u}) \leq 0,\ i = 1, \ldots, N\}.$$

Let the stochastic process $\mathbf{k}(\cdot) = (k_1(\cdot), \ldots, k_{m_c}(\cdot), z_{m+1}(\cdot), \ldots, z_N(\cdot))$, defined on a probability space $(\Omega, \mathcal{F}, P)$, denote the machine-capacity/demand process.

THEOREM 3.3. *The value function is convex and continuous, and satisfies the condition*

$$|v(\mathbf{x},\mathbf{k}) - v(\mathbf{x}',\mathbf{k})| \le C(1 + |\mathbf{x}|^{K_h} + |\mathbf{x}'|^{K_h})|\mathbf{x} - \mathbf{x}'| \tag{3.4}$$

for some positive constant C and all $\mathbf{x}$, $\mathbf{x}' \in S$.

Main Lemma. *Given $\mathbf{x} = (x_1, \dots, x_N)$, $\mathbf{x}' = (x_1', \dots, x_N') \in S$, and $\mathbf{k} \in \mathcal{M}$, let $\mathbf{u}(\cdot) = (u_{ij}(\cdot)) \in \mathcal{A}(\mathbf{x},\mathbf{k})$ and $\mathbf{x}(\cdot)$ be the corresponding trajectory. Then there exists $\mathbf{u}'(\cdot) = (u_{ij}'(\cdot)) \in \mathcal{A}(\mathbf{x},\mathbf{k})$ with the corresponding trajectory $\mathbf{x}'(\cdot)$ such that*

(a) $0 \le u_{ij}'(t) \le u_{ij}(t)$ *for all* $(i,j) \in A$;

(b) For $i = 1, 2, \cdots, N$,

$$\int_0^t \left|U_i^1(\mathbf{u}) - U_i^1(\mathbf{u}')\right| ds = \int_0^t \left(U_i^1(\mathbf{u}) - U_i^1(\mathbf{u}')\right) ds \le \sum_{j=1}^N |x_j - x_j'|, ;$$

(c) $\displaystyle\sum_{n=1}^N |x_n(t) - x_n'(t)| \le \sum_{n=1}^N |x_n - x_n'|.$

4. The Hamilton-Jacobi-Bellman equation

While we do not know if the value function is differentiable, especially at the boundary points of the state constraints, we do know that it is a convex function of the state variable. For convex functions it is convenient to write a dynamic programming equation in terms of *directional derivatives.* So, we first give the notion of these derivatives and some related properties of convex functions.

A function $f(\mathbf{x}), \mathbf{x} \in R^N$, is said to have a *directional derivative* $f_{\mathbf{p}}'$ along direction $\mathbf{p} \in R^N$ if the following limit exists:

$$\lim_{\varepsilon \downarrow 0} \frac{f(\mathbf{x} + \varepsilon \mathbf{p}) - f(\mathbf{x})}{\varepsilon} = f_{\mathbf{p}}'(\mathbf{x}).$$

If a function $f(\mathbf{x})$ is differentiable at $\mathbf{x}$, then $f_{\mathbf{p}}'(\mathbf{x})$ exists for every $\mathbf{p}$ and

$$f_{\mathbf{p}}'(\mathbf{x}) = \langle \nabla f(\mathbf{x}), \mathbf{p} \rangle,$$

where $\nabla f(\mathbf{x})$ is the gradient of $f(\mathbf{x})$ at $\mathbf{x}$ and $\langle \cdot, \cdot \rangle$ is a scalar product. A continuous convex function defined on a convex domain $\sum$ is differentiable almost everywhere, and has a directional derivative along any direction at any inner point of $\sum$ and along any admissible direction (i.e., such direction $\mathbf{p}$ that $\mathbf{x} + \varepsilon\mathbf{p} \in \sum$ for some $\varepsilon > 0$) at any boundary point of $\sum$. Note that $\{A\mathbf{u} : \mathbf{u} \in U(\mathbf{x},\mathbf{k})\}$ is the set of admissible directions at $\mathbf{x}$.

Formally we can write the HJBDD for our problem as:

$$\rho v(\mathbf{x},\mathbf{k}) = \inf_{\mathbf{u} \in U(\mathbf{x},\mathbf{k})} \{v_{A\mathbf{u}}'(\mathbf{x},\mathbf{k}) + h(\mathbf{x},\mathbf{u},\mathbf{k})\} + Qv(\mathbf{x},\mathbf{k}). \tag{4.1}$$

THEOREM 4.1 (Verification). (a) *The value function $v(\mathbf{x},\mathbf{k})$ satisfies equation* (4.1) *for all $\mathbf{x} \in S$.*

(b) *If some continuous convex function $\tilde{v}(\mathbf{x},\mathbf{k})$ satisfies* (4.1) *and the growth condition* (3.4) *with $\mathbf{x}' = 0$, then $\tilde{v}(\mathbf{x},\mathbf{k}) \le v(\mathbf{x},\mathbf{k})$. Moreover, if there exists a feedback control $\mathbf{u}(\mathbf{x},\mathbf{k})$ providing the infimum in* (4.1) *for $\tilde{v}(\mathbf{x},\mathbf{k})$, then $\tilde{v}(\mathbf{x},\mathbf{k}) = v(\mathbf{x},\mathbf{k})$, and $\mathbf{u}(\mathbf{x},\mathbf{k})$ is an optimal feedback control.*

With the notation defined above, it is easy to see that the HJBDD equation associated with $\mathcal{P}(\mathbf{y},\mathbf{k})$ is given as:

$$\rho(\mathbf{k})\hat{v}(\mathbf{x},\mathbf{k}) = \min_{\mathbf{u}\in U(\mathbf{x},\mathbf{k})}[\hat{v}'_{\mathcal{A}\mathbf{u}}(\mathbf{x},\mathbf{k}) + \hat{h}(\mathbf{x},\mathbf{u},\mathbf{k})]. \tag{5.4}$$

PROPOSITION 5.1. *(a) The value function $\hat{v}(\mathbf{y},\mathbf{k})$ of the problem $\mathcal{P}(\mathbf{y},\mathbf{k})$ is convex and satisfies the HJBDD* (5.4).

(b) If $h(\cdot,\cdot,\mathbf{k})$ for some $\mathbf{k}$ is strictly convex in $\mathbf{u}$, then optimal control in $\mathcal{P}(\mathbf{y},\mathbf{k})$ exists, is unique, and can be represented as a feedback control $\hat{\mathbf{u}}(\mathbf{y},\mathbf{k})$.

(c) If some continuous convex function $\tilde{v}(\mathbf{x},\mathbf{k})$ satisfies (4.1) *and the growth condition* (5.4) *with $\mathbf{x}'=0$, then $\tilde{v}(\mathbf{x},\mathbf{k}) \le \hat{v}(\mathbf{x},\mathbf{k})$. Moreover, if there exists a feedback control $\mathbf{u}(\mathbf{x},\mathbf{k})$ providing the infimum in* (5.4) *for $\tilde{v}(\mathbf{x},\mathbf{k})$, then $\tilde{v}(\mathbf{x},\mathbf{k}) = \hat{v}(\mathbf{x},\mathbf{k})$, and $\mathbf{u}(\mathbf{x},\mathbf{k})$ defines the optimal feedback control for $\mathcal{P}(\mathbf{y},\mathbf{k})$.*

(d) Assume, in addition, that $h(\mathbf{x},\mathbf{u})$ is strictly convex in $\mathbf{u}$ for each fixed $\mathbf{x}$. Let $\mathbf{u}^*(\mathbf{x},\mathbf{k})$ denote the minimizer function of the right-hand side of (5.4). *Then*

$$\dot{\mathbf{y}}(t) = \mathcal{A}\mathbf{u}^*(\mathbf{y}(t),\mathbf{k}),\ \mathbf{y}(0)=\mathbf{y} \tag{5.5}$$

has a solution $\mathbf{y}^*(t)$, and $\mathbf{u}(t) = \mathbf{u}^*(\mathbf{y}^*(t),\mathbf{k})$ is the optimal control.

Let $\tau_1,\dots,\tau_n,\dots$ be the sequence of jump times of the process $\mathbf{k}(t)$. Then we can state the following result.

PROPOSITION 5.2. *The value functions $v(\mathbf{x},\mathbf{k})$ and $\hat{v}(\mathbf{x},\mathbf{k})$ coincide and the optimal control for the stochastic problem can be constructed sequentially for $\tau_n < t \le \tau_{n+1}, n=1,\dots,$ from the optimal control for the problems $\mathcal{P}(\mathbf{y},\mathbf{k})$ as follows. If optimal control $\mathbf{u}^*(t)$ and the corresponding optimal trajectory $\mathbf{x}^*(t)$ have already been constructed for $t\le\tau_n$, and if $\hat{\mathbf{u}}(s)$ and $\hat{\mathbf{x}}(s), 0\le s\le\infty$, are the optimal control and the trajectory in the problem $\mathcal{P}(\mathbf{x}^*(\tau_n),\mathbf{k}(\tau_n+0))$, respectively, then $\mathbf{u}^*(t) = \hat{\mathbf{u}}(t-\tau_n)$ and $\mathbf{x}^*(t) = \hat{\mathbf{x}}(t-\tau_n)$ are, respectively, the optimal control and the optimal trajectory in the stochastic problem for $\tau_n < t \le \tau_{n+1}$.*

6. Proof of results

The key result we shall prove in this section is Main Lemma stated in Section 2. The result is an extension of Lemma 1 in Presman, Sethi and Zhang [**PSZ**] (referred to as PSZ hereafter) for the more general problem considered here. Armed with Main Lemma in the place of Lemma 1 of PSZ, it is possible to prove the other results in ways similar to those in PSZ. However, for the convenience of the reader, we shall provide very brief sketches of their proofs later in this section.

In order to prove Main Lemma, let us first consider the deterministic problem $\mathcal{P}(\mathbf{y},\mathbf{k})$ for some initial $\mathbf{y}$ and $\mathbf{k}$ without the objective function, and deterministic functions $\mathbf{y}(t)$ and $\mathbf{u}(t)$, $0\le t<\infty$, which satisfy $\mathbf{y}(0)=\mathbf{y}_0\in S$, $\dot{\mathbf{y}}(t) = \mathcal{A}\mathbf{u}(t)$, and $\mathbf{u}(t)\in U(\mathbf{y}(t),\mathbf{k})$ for the given value of $\mathbf{k}$ (it is easy to see that $\mathbf{y}(t)\in S$ for all $0\le t<\infty$). For any $\mathbf{y}'_0\in S$, we will construct functions $\mathbf{y}'(t)$ and $\mathbf{u}'(t)$ which satisfy $\mathbf{y}'(0)=\mathbf{y}'_0$, $\frac{d}{dt}\mathbf{y}'(t) = \mathcal{A}\mathbf{u}'(t)$, $\mathbf{u}'(t)\in U(\mathbf{y}'(t),\mathbf{k})$ for all $0\le t<\infty$, and $\sum_{i=1}^N[y_i(t)-y'_i(t)]^+$ and $\sum_{i=1}^N[y_i(t)-y'_i(t)]^-$ are nonincreasing in t.

We begin our construction by defining for any $\mathbf{y}'=(y'_1,\dots,y'_N)\in S$ and $\mathbf{u}\in\mathcal{U}$, the disjoint index sets

$$\begin{aligned}\alpha_+(\mathbf{y}',\mathbf{u}) &= \{n : y'_n = H_n, 1\le n\le N,\ U_n^1(\mathbf{u}) - U_n^2(\mathbf{u}) > 0\},\\ \alpha_-(\mathbf{y}',\mathbf{u}) &= \{n : y'_n = L_n, 1\le n\le N-1,\ U_n^1(\mathbf{u}) - U_n^2(\mathbf{u}) \le 0\}.\end{aligned} \tag{6.1}$$

It should be obvious that if $\mathbf{u}$ violates no state constraints, then no modification is required. Furthermore, if using $\mathbf{u}$ would violate only the upper bound (respectively, lower bound) constraints, then $\mathbf{u}^H(\mathbf{y}', \mathbf{u})$ (respectively, $\mathbf{u}^L(\mathbf{y}', \mathbf{u})$) would provide the modified control we are seeking. But in general upper bound constraints may be violated at some buffers and lower bound constraints at some others. Then it seems reasonable that we first modify the control $\mathbf{u}$ according to the backward procedure (6.2) designed to avoid violations of the upper bound constraints, and then further modify the resulting control $\mathbf{u}^H$, in turn, according to the forward procedure (6.3) designed to avoid violations of the lower bound constraints. Thus, we consider the control

$$\mathbf{u}'(\mathbf{y}', \mathbf{u}) = \mathbf{u}^L(\mathbf{y}', \mathbf{u}^H(\mathbf{y}', \mathbf{u})), \tag{6.4}$$

and the resulting (see Lemma 6.3) trajectory obtained from solving the differential equation

$$\frac{d}{dt}\mathbf{y}'(t) = \mathcal{A}\mathbf{u}'(\mathbf{y}'(t), \mathbf{u}(t)), \quad \mathbf{y}'(0) = \mathbf{y}'_0. \tag{6.5}$$

REMARK 6.1. In order to prove their Lemma 1 (corresponding to Main Lemma of this paper), PSZ constructed the stochastic control process $\mathbf{u}'(\cdot) \in \mathcal{A}(\mathbf{x}', \mathbf{k})$, for the new initial condition $\mathbf{x}'$, given $\mathbf{x}, \mathbf{k}$ and $\mathbf{u}(\cdot) \in \mathcal{A}(\mathbf{x}, \mathbf{k})$. Here we construct a simple transformation (6.4), which is local in nature and independent of the machine state $\mathbf{k}$, and which depends only on the new current value (or value at time t) of $\mathbf{y}'$ and the time-t value of the given control $\mathbf{u}$.

We can state and prove the following results concerning the deterministic control defined by (6.4) and the deterministic trajectory defined by (6.5). The first lemma says that the modified control $\mathbf{u}'$ is nonnegative, is no larger than the given control $\mathbf{u}$, and is admissible at any point in S. The second says that the corresponding trajectory $\mathbf{y}'(t)$ exists, is unique, and stays within the required upper and lower bounds for $t \in [0, \infty)$. Intuitively, it is not difficult to see why these results should follow from our construction. The second lemma also says that the deterministic trajectory $\mathbf{y}(\cdot)$, emanating from the given initial condition $\mathbf{y}$ and the control $\mathbf{u}(\cdot)$, and the new trajectory $\mathbf{y}'(\cdot)$ does not stray too far away from one another from where they begin at time zero.

LEMMA 6.2. *If* $\mathbf{u} \in U(\mathbf{y}', \mathbf{k})$, *then*

(i) $0 \le u'_{ij}(\mathbf{y}', \mathbf{u}) \le u_{ij}$, $(i,j) \in A$;

(ii) $\mathbf{u}'(\mathbf{y}', \mathbf{u}) \in U(\mathbf{y}', \mathbf{k})$.

PROOF. In order to prove (i), it suffices to show the following:

(H_1) $0 \le u^H_{ij}(\mathbf{y}', \mathbf{u}) \le u_{ij}$, $(i,j) \in A$, and

(L_1) $0 \le u^L_{ij}(\mathbf{y}', \mathbf{u}) \le u_{ij}$, $(i,j) \in A$.

To prove (H_1), we use the backward induction argument (on the columns of (u_{ij})) in view of $u^H_{i,N+1}(\mathbf{y}', \mathbf{u}) = u_{i,N+1}, m+1 \le i \le N$, with the knowledge that $U^2_n(\mathbf{u}) \le U^1_n(\mathbf{u})$ if $n \in \alpha_+(\mathbf{y}, \mathbf{u}')$, and $0 \le (a+b)^+ \le a$ for

$$a = U^1_n(\mathbf{u}) \ge 0,$$

$$b = \sum_{i=n+1}^{N} u^H_{ni}(\mathbf{y}', \mathbf{u}) - U^2_n(\mathbf{u}) = U^H_n(\mathbf{y}', \mathbf{u}) - U^2_n(\mathbf{u}) \le 0.$$

We can prove (L_1) analogously.

We now consider the sum of the negative differences. It is easy to see that for the piecewise-constant, right-continuous control $\mathbf{u}(\cdot)$, a fixed t, and a sufficiently small value of s, we have

$$\begin{aligned}\sum_{n=1}^{N}[y_n(t+s)-y'_n(t+s)]^- &= \sum_{n\in\alpha_+(\mathbf{y}(t),\mathbf{y}'(t),\mathbf{u}(t))}[y'_n(t+s)-y_n(t+s)] \\ &= \sum_{n=1}^{N}[y_n(t)-y'_n(t)]^- \\ &\quad -s\sum_{n\in\alpha_+(\mathbf{y}(t),\mathbf{y}'(t),\mathbf{u}(t))}(\mathcal{A}(\mathbf{u}(t)-\mathbf{u}'(\mathbf{y}'(t),\mathbf{u}(t))))_n.\end{aligned}$$

It is clear that for $\sum_{n=1}^{N}[y_n(t+s)-y'_n(t+s)]^-$ to be nonincreasing, it is enough that for any $\mathbf{y}$, $\mathbf{y}'$, and $\mathbf{u}$, we have

$$\sum_{n\in\alpha_+(\mathbf{y},\mathbf{y}',\mathbf{u})}(\mathcal{A}(\mathbf{u}-\mathbf{u}'(\mathbf{y}',\mathbf{u})))_n\geq 0. \tag{6.8}$$

To show (6.8), we need to explore the relationship between the index sets defined in (6.1) and (6.6). Since $U_n^1(\mathbf{u})-U_n^2(\mathbf{u})>0$ and $y'_n=H_n$ imply $y_n<H_n$, it is immediate that $\alpha_+(\mathbf{y}',\mathbf{u})\subset\alpha_+(\mathbf{y},\mathbf{y}',\mathbf{u})$. On the other hand, $U_n^1(\mathbf{u})-U_n^2(\mathbf{u})\leq 0$ and $y'_n=L_n$ imply $y_n\geq L_n$ and $(\mathcal{A}(\mathbf{u}-\mathbf{u}'(\mathbf{y}',\mathbf{u})))_n=0$ in case $y_n=y'_n=L_n$. Thus, $\alpha_-(\mathbf{y}',\mathbf{u})\subset\alpha_-(\mathbf{y},\mathbf{y}',\mathbf{u})$.

We can now prove (6.8). We rewrite (6.7) as

$$\begin{aligned}&(\mathcal{A}(\mathbf{u}-\mathbf{u}'(\mathbf{y}',\mathbf{u})))_n \\ &\qquad=(\mathcal{A}(\mathbf{u}-\mathbf{u}^H(\mathbf{y}',\mathbf{u})))_n+(\mathcal{A}(\mathbf{u}^H(\mathbf{y}',\mathbf{u})-\mathbf{u}^L(\mathbf{y}',\mathbf{u}^H(\mathbf{y}',\mathbf{u}))))_n.\end{aligned} \tag{6.9}$$

From the definition of $\mathbf{u}^H(\mathbf{y}',\mathbf{u})$ and the remark just after (6.2), we can easily conclude

- $(\mathcal{A}(\mathbf{u}-\mathbf{u}^H(\mathbf{y}',\mathbf{u})))_n=0$ if $n\notin\alpha_+(\mathbf{y}',\mathbf{u})$ and $U_n^1(\mathbf{u})-U_n^2(\mathbf{u})\geq 0$, and
- $(\mathcal{A}(\mathbf{u}-\mathbf{u}^H(\mathbf{y}',\mathbf{u})))_n\leq 0$ if $n\notin\alpha_+(\mathbf{y}',\mathbf{u})$ and $U_n^1(\mathbf{u})-U_n^2(\mathbf{u})\leq 0$.

Therefore,

$$(\mathcal{A}(\mathbf{u}-\mathbf{u}^H(\mathbf{y}',\mathbf{u})))_n\leq 0,\qquad \forall n\notin\alpha_+(\mathbf{y}',\mathbf{u}). \tag{6.10}$$

Moreover, by (6.7), (6.2), and Lemma 6.2, we have

$$\sum_{n=1}^{N}(\mathcal{A}(\mathbf{u}-\mathbf{u}^H(\mathbf{y}',\mathbf{u})))_n=\sum_{i=0}^{m-1}u_{0i}-\sum_{i=0}^{m-1}u_{0i}^H(\mathbf{y}',\mathbf{u})\geq 0. \tag{6.11}$$

Inequalities (6.10) and (6.11) together with the fact that $\alpha_+(\mathbf{y}',\mathbf{u})\subset\alpha_+(\mathbf{y},\mathbf{y}',\mathbf{u})$ allow us to conclude

$$\begin{aligned}&\sum_{n\in\alpha_+(\mathbf{y},\mathbf{y}',\mathbf{u})}(\mathcal{A}(\mathbf{u}-\mathbf{u}^H(\mathbf{y}',\mathbf{u})))_n \\ &\qquad=\sum_{n=1}^{N}(\mathcal{A}(\mathbf{u}-\mathbf{u}^H(\mathbf{y}',\mathbf{u})))_n-\sum_{n\notin\alpha_+(\mathbf{y},\mathbf{y}',\mathbf{u})}(\mathcal{A}(\mathbf{u}-\mathbf{u}^H(\mathbf{y}',\mathbf{u})))_n \\ &\qquad\geq 0.\end{aligned} \tag{6.12}$$

Similarly, the following can be verified easily from the definition of $\mathbf{u}^L(\mathbf{y}',\mathbf{u}^H(\mathbf{y}',\mathbf{u}))$:

Proof of Main Lemma. For given $\mathbf{u}(\cdot) \in \mathcal{A}(\mathbf{x}, \mathbf{k})$ and the new initial condition $\mathbf{x}' \in S$, we can construct the continuous process $\mathbf{x}'(t)$ satisfying

$$\frac{d\mathbf{x}'(t)}{dt} = \mathcal{A}\mathbf{u}'(\mathbf{x}'(t), \mathbf{u}(t)), \qquad \mathbf{x}'(0) = \mathbf{x}',$$

by applying the existence statement of Lemma 6.3 between successive jumps of the process $\mathbf{k}(\cdot)$. It follows from the above construction that the properties of controls and trajectories specified in Lemmas 6.2 and 6.3 hold also for the processes $\mathbf{x}(\cdot)$, $\mathbf{u}(\cdot)$, $\mathbf{x}'(\cdot)$ and $\mathbf{u}'(\cdot)$. This proves statements (a) and (c) of Main Lemma.

In order to prove (b), we know from (2.4) that

$$\sum_{i=n}^{N} \frac{d}{dt}(x_i(t) - x_i'(t)) = \sum_{i=n}^{N} \sum_{j=0}^{(i-1)\wedge m} (u_{ji}(t) - u_{ji}'(t)). \tag{6.16}$$

Integrating from 0 to t and using the results of Lemmas 6.2 and 6.3, we have

$$\begin{aligned}
\int_0^t & \left| \sum_{i=0}^{(n-1)\wedge m} u_{in}(s) - \sum_{i=0}^{(n-1)\wedge m} u_{in}'(s) \right| ds \\
&\le \sum_{i=n}^{N} \left| \int_0^t \sum_{j=0}^{(i-1)\wedge m} (u_{ji}(s) - u_{ji}'(s)) \right| ds \\
&= \sum_{i=n}^{N} (x_i(t) - x_i'(t)) - \sum_{i=n}^{N} (x_i - x_i') \\
&\le \sum_{i=1}^{N} (x_i(t) - x_i'(t))^+ + \sum_{i=1}^{N} (x_i - x_i')^- \\
&\le \sum_{i=1}^{N} (x_i - x_i')^+ + \sum_{i=1}^{N} (x_i - x_i')^- = \sum_{i=1}^{N} |x_i - x_i'|.
\end{aligned}$$

This proves statement (b), and completes the proof of Main Lemma. □

As mentioned earlier, given Main Lemma in place of Lemma 1 in PSZ, Theorems 3.3, 5.1 and 5.2 and Theorems 3.2 and 4.1 can be proved essentially in the same way as Propositions 1, 2, and 3 and Theorems 1 and 2 in PSZ, respectively. For the reader's convenience, we give brief sketches of these results. Here, Proposition 5.1 follow from (b) and (c) of Main Lemma and Assumption (A2). In Proposition 5.2, the results that the function $\hat{h}(\cdot, \cdot, \mathbf{k})$ is convex, continuous, and satisfies the Lipschitz property (3.4) follow easily from Proposition 4.1. The remainder of the proof is the same as the proof of Proposition 2 in PSZ. The proof of Proposition 5.2 is the same as the proof of Proposition 3 in PSZ. Finally, one can easily see that Theorems 3.2 and 4.1 follow from Propositions 5.1 and 5.2; see Sethi and Zhang [**SZ**, Chapter 4] for details.

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